

A REPORT ON INTERNSHIP AT STARC LAB, IIT INDORE



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Foreground removal of 21 cm cosmological signals using GPR

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Abstract

This report explores the use of Gaussian Process Regression (GPR) for foreground removal in the context of 21 cm cosmology. The 21 cm signal, originating from the hyperfine transition of neutral hydrogen, is a powerful probe of the Universe's structure during the Epoch of Reionization (EoR). However, observations of this faint signal are dominated by bright astrophysical foregrounds, including Galactic synchrotron and free-free emissions, extragalactic sources, and the Cosmic Microwave Background (CMB). The work leverages GPR, a non-parametric Bayesian method, to model the foregrounds as smooth correlated random fields and distinguish them from the oscillatory 21 cm signal. The methodology is applied to synthetic datasets, demonstrating its ability to recover the signal while preserving its fluctuations. Despite its robustness, challenges such as dependence on kernel choice, computational cost, and handling polarized emissions are noted. The study highlights the need for improved methods in foreground removal, laying the groundwork for advancing 21 cm intensity mapping techniques.

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1 Introduction

1.1 Phases of Cosmic Evolution

Since consciousness dawned upon the ancestors of our civilisation, they have questioned their place in this world and their surroundings. Though the study of the solar system was among the oldest of sciences, the search soon extended beyond and we have, today, a whole branch dedicated to the study of the universe and its evolution,- Cosmology. The evolution of the universe began as early as 14 billion years ago and will extend to the far future. The whole of evolution can be divided into six main stages, with epochs in them:.

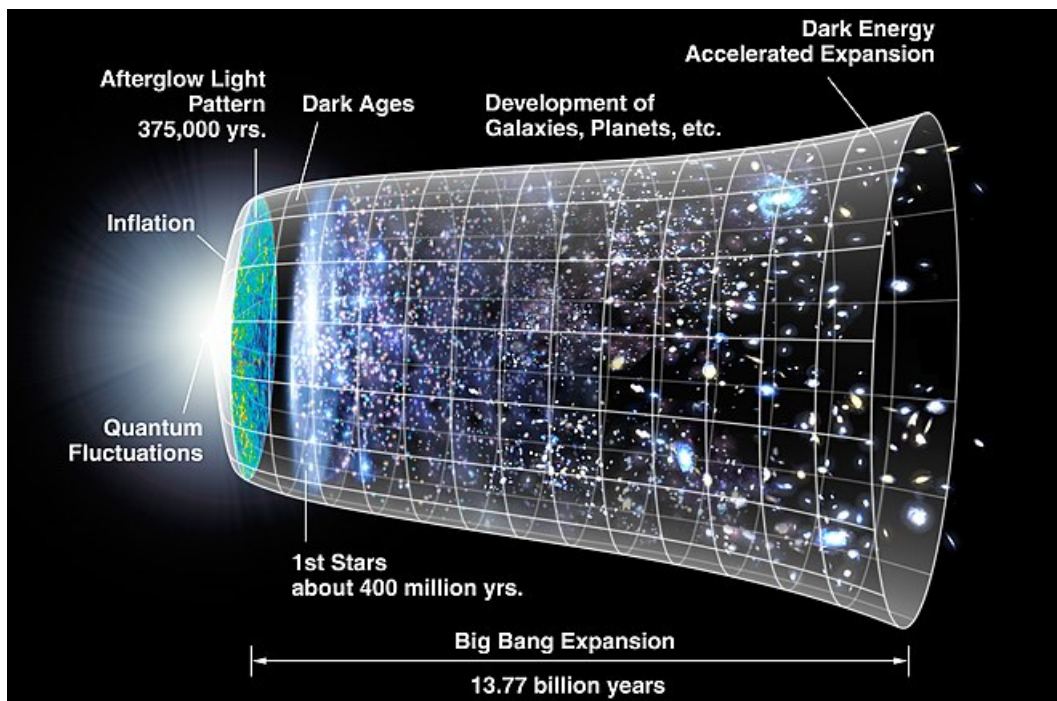


Fig. 1: Evolution timescale of the Universe

1. The **Primordial Era** has a time frame of less than 10^{-6} seconds after the Big Bang. It consists of the **Planck Epoch** where gravity separated from the other fundamental forces, the **Grand Unification epoch** where the strong force separated from the electroweak force, the Inflationary epoch where the universe smoothed due to exponential expansion and the Quark epoch where the quark-gluon plasma formed.
2. The **Radiation Era**, from 10^{-6} seconds to 50000 years after the Big Bang. It has the **Hadron epoch** where quarks combined, the **Lepton epoch** with annihilation processes and the **Photon epoch**, forming the cosmic plasma.

3. The **Matter Era**, from 50000 billion years to 1 billion years after the big bang. During this time, the gravitational effects of matter became more dominant over the radiation pressure, forming Baryonic and Dark matter. The main epochs are **Recombination epoch** where the cosmic microwave background formed, the **Dark Ages** where the universe was filled with neutral hydrogen and devoid of any celestial bodies and the **Cosmic Dawn** when the first stars and galaxies formed.
4. The **Reionization Era**, marking the transition from Dark ages, when the first stars and galaxies ionised the neutral hydrogen in the IGM. This is the period of the **Epoch of Reionization (EoR)**.
5. The **Stellar era**, right from the EoR to the present day, when stars and planets formation peaked, heavier elements formed.
6. The **Dark Energy era**, after 5 billion years of the big bang, where dark energy will dominate the universe's density leading to severe accelerated expansion, with decline in star formation and gradual "big freeze" or heat death of the universe.

These phases have several unique ways of detection and study, such as detecting fluctuations in CMB, from the various cosmic signals, using GW detectors or reproducing the particle phenomenology. The cosmic signals stretch across the entire electromagnetic range with some wavelengths more suitable than others for certain observations.

1.2 21 cm line

The **Epoch of Reionization** marks the transition from the dark ages to the stellar ages. During this time, the neutral hydrogen filling the cosmos was ionized by the first stars and galaxies. The hydrogen atom undergoes a spin flip transition, which means the direction of the electron's spin is reversed relative to the spin of the proton. This is a quantum state change between the two hyperfine levels of the hydrogen 1s ground state. The electromagnetic radiation produced has a wavelength of 21 cm in vacuum and hence we call this spectral line as "21 cm signal". This is of course a radio signal. The 21 cm line is the only source to probe the EoR. First, by mapping the intensity of redshifted 21 cm radiation it can, in principle, provide a very precise picture of the matter power spectrum in the period after recombination. Second, it can provide

a picture of how the universe was re-ionized, as neutral hydrogen which has been ionized by radiation from stars or quasars will appear as holes in the 21 cm background.

But these observations are difficult to make and analyse because of various unwanted sources.

2 The Astrophysical Problem

2.1 What are foregrounds?

Following reionization, most neutral hydrogen in the interstellar medium was ionized, leaving only a small fraction within self-shielding galaxies. As a result, HI effectively traces the matter distribution in the Universe. Instead of examining individual galaxies, intensity mapping (IM) measures the combined HI emission from multiple galaxies to quantify the structure in a region of the sky. This method enables the mapping of vast sky areas more efficiently than traditional spectroscopic galaxy surveys. But the major problem arises due to the astrophysical foregrounds that plague the 21 cm signal. In order to access the HI, we need to remove these signals from our data.

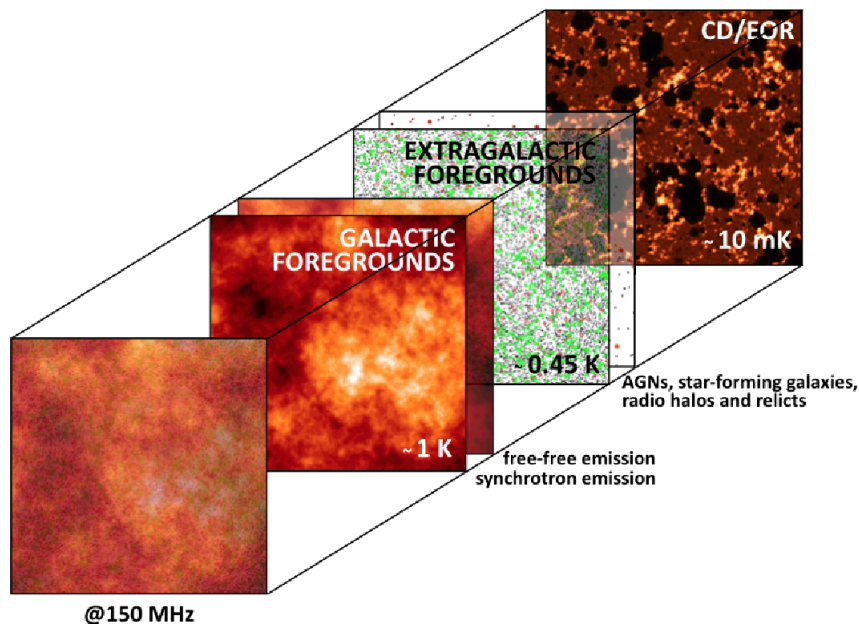


Fig. 2: Sources of foreground signals

The primary sources of these foreground signals are:

- Our galaxy: Most of the diffuse emission comes from our galaxy, over 75%, due to synchrotron processes.
- Other Galactic Sources: They are caused mainly by synchrotron radiations, bremsstrahlung in the warm ionized medium and anomalous microwave emission (AME) from spinning

dust grains.

- Extragalactic sources: Point sources such as AGNs or star forming galaxies emitting in radio bands and the CMB at higher frequencies are the major sources.

2.2 Removal principles and methods

We exploit the difference in properties in the foregrounds and the HI signal to separate the two. Mathematically, the combined foreground is several orders of magnitude brighter than the HI signal in the same radio frequency. However, they follow a power law and vary slowly in frequency, so has a smooth spectral pattern, while the HI signal is tracing the large-scale structure of the Universe and so is not smooth in frequency, rather shows a very oscillatory behavior. We can use several methods to separate these signals, the popular ones being Principal Component Analysis (PCA), Independent Component Analysis (ICA), Polynomial fitting and Gaussian Process Regression (GPR).

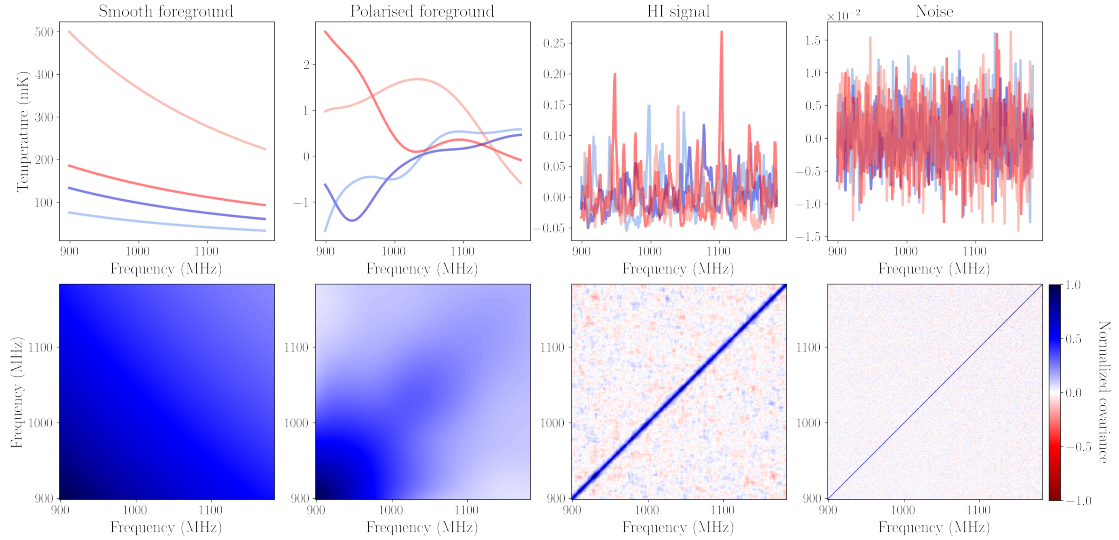


Fig. 3: Components of the data

PCA is one of the most common methods, where the covariance matrix of the data is calculated followed by its eigenvalue decomposition, then the first few eigenmodes, assumed to correspond to the smooth foreground, are subtracted. Though this is one of the best methods, the frequency variation of the foregrounds are not smooth throughout the sky patch, so the number

of eigenmodes subtracted may sometimes lead to loss of HI data as well.
We have, in our work, used the Gaussian Process Regression.

3 What I did

Gaussian Process Regression method assumes the foreground to be a correlated random field with smooth spectral behavior. Our data ($\mathbf{d}$) basically has 3 components: smooth foreground data ($\mathbf{f}_{fg}$), HI signal ($\mathbf{f}_{21}$) and gaussian noise ($\mathbf{f}_n$). Our target is to figure out the foreground data. Each component of our data is a Gaussian process. Now the foreground is treated to have a smooth covariance kernel (K_{fg}) and a model is designed. This model predicts the foreground component by maximising the marginal likelihood of the data, the hyperparameters optimised and finally after fitting, the foregrounds are subtracted to give the HI signal and noise.

$$\mathbf{d} = \mathbf{f}_{fg} + \mathbf{f}_{21} + \mathbf{n}$$

$$f(\boldsymbol{\nu}) \sim N(m(\boldsymbol{\nu}), k(\boldsymbol{\nu}, \boldsymbol{\nu})) \text{ where } K_{fg} = k(\boldsymbol{\nu}, \boldsymbol{\nu})$$

Since our data has 3 separate components, we can write: $K = K_{fg} + K_{21} + K_n$ If we have a data $\mathbf{d}$ over frequency $\boldsymbol{\nu}$, we can obtain its joint probability with another set of frequencies $\boldsymbol{\nu}'$ as:

$$\begin{bmatrix} \mathbf{d} \\ \mathbf{d}' \end{bmatrix} = N \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} k(\boldsymbol{\nu}, \boldsymbol{\nu}) & k(\boldsymbol{\nu}, \boldsymbol{\nu}') \\ k(\boldsymbol{\nu}', \boldsymbol{\nu}) & k(\boldsymbol{\nu}', \boldsymbol{\nu}') \end{bmatrix} \right)$$

Modeling the foregrounds $\mathbf{f}_{fg}$, we get

$$\begin{bmatrix} \mathbf{d} \\ \mathbf{f}_{fg} \end{bmatrix} = N \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} K_{fg} + K_{21} + K_n & K_{fg} \\ K_{fg} & K_{fg} \end{bmatrix} \right)$$

Our foreground is basically $\mathbf{f}_{fg} \sim N(E[\mathbf{f}_{fg}], cov[\mathbf{f}_{fg}])$. The expectation value is the prediction of the foreground signal and the covariance matrix is the uncertainty. Finally we obtain the residual data as $\mathbf{r} = \mathbf{d} - E[\mathbf{f}_{fg}]$.

The link to my work is [here](#).

The dataset I have used has dimensions (128, 128, 142). Each value corresponds to one observation.

4 Future Prospects

We have used Gaussian Process Regression which is quite a robust method and preserves the fluctuations at low frequencies. There are still some limitations of this method. The foreground predictions depend on the choice of the right kernel, which is perhaps the biggest drawback. Polarization reduces the smoothness of the data making it difficult to apply this property and thus, smooth foreground is a very idealised technique. Noise itself has a very oscillatory pattern like the HI signals, and thus we cannot eliminate it using this method. Most importantly, this is a very computationally expensive technique for large datasets. This demands the need of better methods of foreground removal, which happens to be an active area of work now.

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